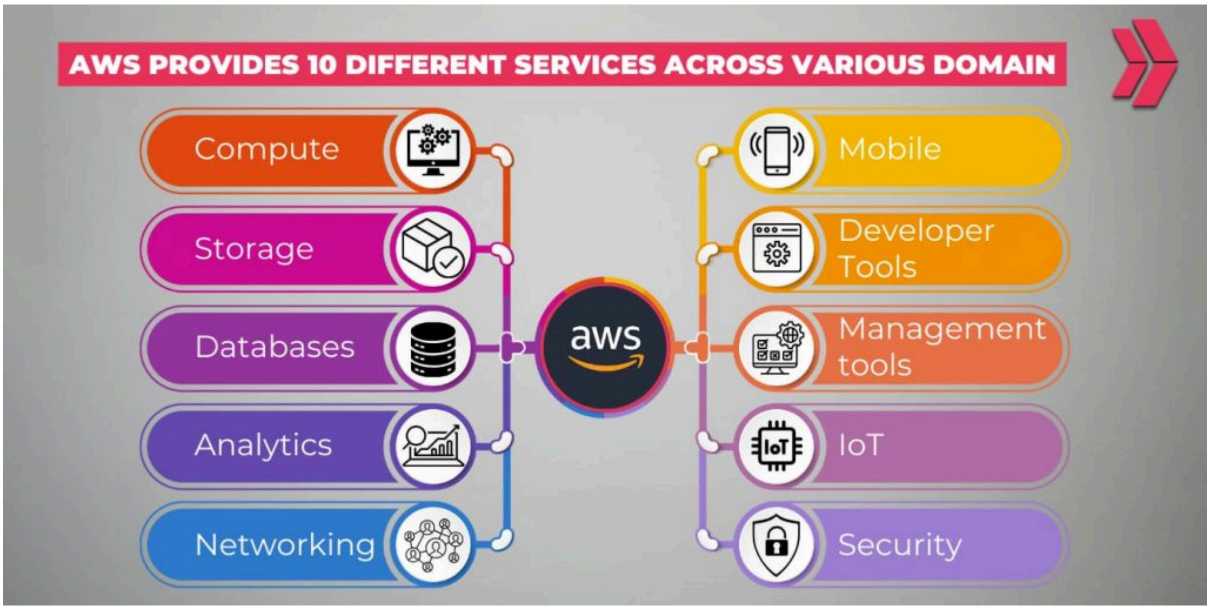
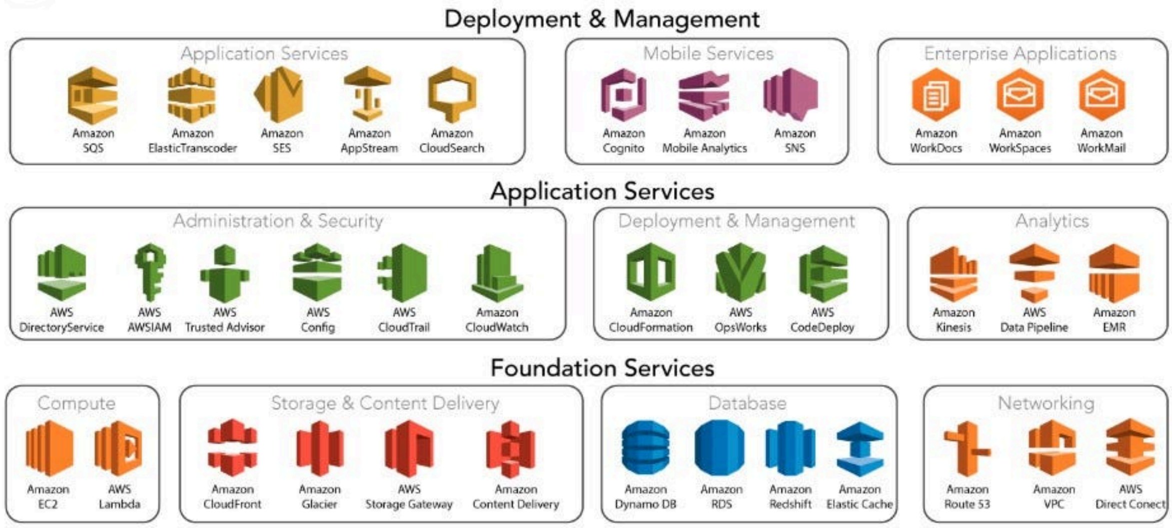
AWS Cloud, also known as Amazon Web Services, is a comprehensive cloud computing platform provided by Amazon. AWS Cloud offers over 200 full-featured services from leading data centers worldwide, enabling businesses, developers, and individuals to harness the power of cloud technology without the need to invest in building and maintaining complex infrastructure.

Thanks to Amazon Web Services, businesses can leverage cloud technology to reduce data storage costs, enhance efficiency, and foster innovation. Beyond technical benefits, AWS Cloud helps turn ideas into reality quickly – unlocking new opportunities and enabling businesses to address challenges in an increasingly complex digital world.



Types of AWS Cloud Services

Currently, AWS Cloud offers three main types of cloud computing – each providing different levels of control, management, and flexibility, making it easy for customers to choose the service that best suits their needs:



1. Platform as a Service (PaaS)

A PaaS (Platform-as-a-Service) platform eliminates the need to manage your organization's underlying infrastructure (including operating systems and hardware), allowing you to focus on deploying and managing existing applications.

This allows you to work more efficiently without worrying about issues such as software maintenance, capacity planning, bug fixing, resource procurement, or any other tasks related to application operation.

2. Software as a Service (SaaS)

SaaS (Software-as-a-Service) software provides complete products that are operated and managed by leading vendors. Furthermore, most of these applications are designed for end users.

With SaaS, you don't need to worry about managing the underlying infrastructure or maintaining the service; you only need to consider how to use the software to store data and grow your business.

A common example of SaaS is web-based email: You can send and receive emails without adding extra features to email or maintaining the operating system and servers for the email program.

3. Infrastructure as a Service (IaaS)

IaaS (Infrastructure-as-a-Service) cloud services provide the basic building blocks for an IT cloud platform with features such as access to computing (dedicated or virtual hardware) and data storage capacity.

IaaS offers users the highest level of flexibility in using and managing IT resources – most closely resembling the IT resources commonly used by many developers and IT departments today.

Who Should Use AWS Cloud Services?

Every business and organization, regardless of industry or size, should utilize AWS cloud computing services for data backup, data recovery, virtual desktops, development, and analytics. Organizations of all types, sizes, and industries are using cloud services for a variety of use cases, including:Data backup, disaster recovery, email, virtual desktops, big data development and analysis, and customer-interactive web applications.

For example:

- Healthcare companies use AWS Cloud services to build and develop the most optimal treatment methods based on each patient's condition.
- Financial services companies use AWS services to enhance their ability to detect and prevent fraud in real time.
- Video game developers utilize AWS Cloud services to create engaging online games for millions of customers worldwide.

Key Features of AWS Cloud Services

Here are some key features of AWS Cloud services to give you a clearer understanding of the platform before using it:

1. Store on Amazon S3 (Amazon Simple Storage)

Amazon S3 is a platform for storing and retrieving all data at any time on demand. Amazon S3 offers top-tier security, high performance and compatibility, and unlimited scalability at an extremely affordable cost.

Amazon S3 makes it easy for users to store, retrieve, and analyze large amounts of data quickly whenever needed. This saves you the cost of purchasing hardware and unnecessary storage space. Instead, you only need to buy bandwidth and memory based on your actual usage requirements.

This service provides users with various management methods, access options, and REST and SOAP web service APIs for efficient data storage, retrieval, and management.

The stored data will be displayed as objects in folders called buckets. Within each bucket, you can perform operations such as creating, deleting, and displaying topics. However, to save topics to the database, you need to upload files to the designated bucket.

The steps are quite simple and are as follows:

- Step 1: Log in to the website aws.amazon.com=> Select S3 located under the "Services" section.
- Step 2: The S3 dashboard will display all available buckets and options to create a new bucket => select "Create Bucket". Once the new bucket is created, you can see its properties on the right.
- Step 3: Click on the newly created bucket to access the files it contains => select the "Upload Button" to upload the files.
- Step 4: You can set access permissions for the file by selecting the file => choosing "Make public". The file can now be accessed using the REST API.

2. Amazon EC2 (Elastic Computing Amazon) service

AWS offers EC2 services, allowing users to easily deploy applications without investing in hardware. Additionally, users can adjust storage capacity directly on the platform.

EC2 AmazonEC2 helps users create multiple server resources according to their specific needs, enhancing security and network configuration. You can use EC2 to scale up and down to support faster workload processing with the following key features:

- Instances are the computations performed by a virtual environment.
- Amazon Machine Images are pre-configured templates for instances in terms of memory, storage capacity, system power, and CPU.
- Secure your login information by storing the public key via AWS key pair. Users will store the private key.
- Temporary data is deleted when Instances are interrupted or paused; these are called Instance Store Volumes.
- Amazon Elastic Block Store, or Amazon EBS volumes, is responsible for providing persistent storage capacity for user data.

- Resources are stored in various locations such as Amazon EBS volumes or instances, known as Regions and Availability Zones.
- Users can assign metadata to Amazon EC2.
- Elastic IP Addresses is the name given to static IPv4 addresses.
- Firewall services allow users to customize the IP address ranges, protocols, and ports that can be associated with the instances they are using.
- It is possible to isolate the virtual network from the rest of the AWS Cloud system and connect it to a user's private network – known as Virtual Private Clouds.

Why is AWS Cloud the Leading Cloud Solution Today?

AWS Cloud has become the leading cloud solution, chosen and used by many customers today thanks to a number of outstanding features:

1. Fully featured

AWS is a leading cloud service provider, offering users not only basic solutions like computing, storage, and data management, but also applying advanced technologies such as machine learning, artificial intelligence, and IoT. This makes it easier for businesses to migrate applications to the cloud environment quickly, efficiently, and cost-effectively.

In addition, AWS offers many specialized database options optimized for a wide variety of applications. This allows your business to choose the option that best suits your budget and performance needs.

2. Proven operational expertise

With over 20 years of experience in the cloud computing field, AWS Cloud provides users with the best storage solutions and helps them stand firm against industry challenges. AWS has become a symbol of stability, security, and high performance – a trusted destination for the most critical applications of many organizations.

With its global operations and service to millions of customers, AWS is a testament to its leadership not only in technology but also in its ability to understand and meet the diverse needs of its customers.

Combined with its enormous scale and influence, this makes AWS one of the world's leading and most trusted cloud service providers.

3. The pace of innovation in line with trends.

With AWS's continuous innovation, you can always be at the forefront of using new technologies to optimize your business and drive sustainable growth.

A prime example is AWS's introduction of AWS Lambda in 2014, a pioneering service in the serverless computing space. This allows developers to run code without managing servers, optimizing the development process and creating more flexible and efficient applications.

Furthermore, AWS has developed Amazon SageMaker for those without prior machine learning experience. This provides developers and scientists with the opportunity to more easily access and apply machine learning to their daily projects.

4. Top-tier security

AWS Cloud is designed with the mission of becoming the world's leading secure and flexible cloud computing environment. AWS's core infrastructure can effectively meet the high security needs of banks, the military, and other organizations.

This commitment is demonstrated through more than 300 unique security features and services, along with rigorous compliance and governance. AWS currently supports 143 security standards and compliance certifications, ensuring that our customers' cloud environments meet the most stringent requirements.

In particular, all 117 AWS services that provide data storage have data encryption capabilities, ensuring the privacy and security of information throughout the migration and storage process.

5. The largest community of customers and partners.

AWS has a vast and diverse connected community with millions of customers and thousands of partners worldwide. AWS customers come from every industry and every size, from startups to multinational corporations – they use AWS Cloud services to deploy every type of application and service imaginable on the platform.

The AWS Partner Network (APN) comprises thousands of AWS cloud computing system integrators and tens of thousands of independent software vendors (ISVs) that have integrated their technology to operate on AWS. This provides customers with a flexible choice of solutions and applications developed by a diverse and extensive network of partners.

What benefits does AWS Cloud service offer customers?

Based on the outstanding features of AWS Cloud, it can be concluded that AWS cloud computing and storage services offer many important benefits to customers, specifically:

1. Flexible scalability

With AWS cloud computing, businesses don't need to use excessive resources to handle future business operations. Instead of investing upfront in large infrastructure, you only need to purchase the resources you actually need. This allows you to flexibly scale resources up or down to precisely meet your business needs.

2. Cost savings

AWS Cloud services allow users to manage IT costs more effectively. Instead of paying a large fixed fee for data centers and physical servers, you only pay for the IT resources you actually need. This optimizes costs and allows for more flexible budget allocation in each case.

3. Excellent growth rate

AWS Cloud provides users with access to a range of cutting-edge technologies for rapid data acquisition to drive innovation and growth. This includes infrastructure services such as computing, storage, and databases, as well as the Internet of Things (IoT), machine learning, data warehousing, and analytics.

Cloud services allow you to deploy technology services more efficiently from concept to completion, helping you create a flexible environment for experimentation and finding new ideas to enhance customer experience, driving business growth and success.

4. Deploy globally in minutes.

With AWS Cloud, you can expand your presence to new geographic areas and deploy applications globally in minutes.

For example, AWS has a global infrastructure, allowing users to deploy their applications in multiple locations with just a few clicks. By placing applications closer to end users, you can reduce latency and enhance the user experience. This creates global connectivity and improves interaction with customers worldwide.

AWS Cloud Service Deployment Model

AWS Cloud services offer the following common deployment models:

1. Cloud model

Cloud-based applications are entirely deployed and operate in a cloud environment. They can be developed directly in the cloud or migrated from existing infrastructure to the cloud to take advantage of the benefits of cloud computing.

These applications can be designed on lower-level infrastructure layers or leverage higher-level services that reduce the need for management, architectural design, and scaling of the application's core infrastructure, allowing for a focus on developing the application's core functionality.

2. Hybrid model

Hybrid deployment is a method of combining infrastructure and applications between cloud resources and existing non-cloud resources.

The most common form of hybrid deployment is a combination of existing on-premises and cloud infrastructure, aiming to extend and enhance an organization's infrastructure into the cloud environment while maintaining connectivity between cloud resources and internal systems.

3. On-site model

Deploying resources on-premises with the support of virtualization technology and resource management tools is sometimes also referred to as a “private cloud.” While this form doesn’t offer many of the benefits of cloud computing, it is favored for its ability to provide specialized resources.

Typically, this model combines traditional IT infrastructure with the use of virtualization and application management technologies to improve resource utilization efficiency.

Reliable and Affordable Cloud Storage Services in Vietnam

Renova Cloud proudly offers leading cloud storage services in Vietnam – providing secure, flexible, and cost-effective cloud storage solutions for businesses.

Based on advanced technology, Renova Cloud has helped many businesses leverage the benefits of cloud computing, including:VietceteraVietcetera, a rapidly growing media startup trusted by both domestic and international investors, currently hosts and operates its services on Google Cloud Platform (GCP). However, seeking a change to find the most suitable service, the company decided to migrate its system to AWS. Renova Cloud, a partner of both AWS and GCP, was honored to be selected by Vietcetera to collaborate on this migration process.

Renova Cloud's cloud storage serviceSuitable for businesses of all sizes, from startups to large organizations, and secured with industry-leading security measures. Renova Cloud's professional technical team is always ready to support any request, ensuring your cloud storage service runs smoothly and efficiently.